Attention-enhanced Swin Transformer For Multi-label Pneumonia And TB Detection With Severity Assessment

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Abstract—The quick detection of diseases of the lung is critical in ensuring that there is a reduction in mortality and clinical decision-making. This is the ability to identify and distinguish various lung diseases using one chest x-ray image. This study provides the solution to this problem through the use of attention-enhanced multi-labelled chest X-ray image classification method with severity evaluation. The model is trained using pre-existing publicly available datasets for chest focussed X-ray images of normal, TB and Pneumonia. This particular framework is based on a Swin Transformer-based architecture with an attention mechanism to pay attention to disease-relevant lung areas, which enhances the diagnostic accuracy and interpretability. Besides this, there is a severity assessment module that estimates the extent of the disease, based on the areas of the lung which are activated. The next step in the work could be the expansion of the dataset and the system verification in the real clinical setting.

Keywords—Multi-Class Classification, Chest X-ray Images, Attention Mechanism, Swin Transformer, Severity Assessment.

I. Introduction

Much of the respiratory mortality that can be prevented is attributed to pneumonia and tuberculosis (TB). The traditional clinical diagnosis is based on the skills of radiologists and manual analysis of the images of the chest X-rays, which can be rather time-consuming and subjective. Also, a patient can have overlapping radiographic patterns or different levels of severity, and thus single-class prediction models cannot be simple to use. Recent advancements in the sphere of deep learning, specifically Vision Transformers and attention mechanisms, have demonstrated a high potential in medical image analysis. We incorporate our model with Swin Transformer-based architecture combined with attention mechanism to enhance the feature representation and localization of disease-specific regions. It is designed to classify the normal, Pneumonia, and Tuberculosis cases and concentrate on the pathology-related features thereby improving the effectiveness, interpretability, and precision. The chest X-ray datasets that are publicly available with labelled images of Pneumonia, Normal and TB cases are used in model training. The pre-processing operations of resizing, normalizing and data augmentation are performed to enhance robustness and generalization. Besides classification, the system has a severity assessment component that estimates the severity of infection in the parts of the lungs, and Grad-CAM visualization is taken to

give explainability by highlighting the regions that affect the model prediction. The following parts of the paper will be organized in such a way that they will contain a literature review, model architecture, preprocessing, and training approach, experimental findings, performance indicators, and discussion of clinical implication to enhance early detection and patient outcomes.

II. Related Works

K. S. Charan et al. [1] proposed a multi-class lung disease prediction system based on transfer learning, using textural features generated using fusion data, and this study showed that hybrid feature extraction methods are more effective in enhancing classification accuracy, which methodologically justifies feature-oriented learning approaches in our transformer-based system.

M. M. Kabir et al. [2] have given a knowledge based distillation model to identify TB, Pneumonia and Covid-19 using radiographs of chest, focusing on effective model compression and multi-disease recognition, which fits our goal of effective multi-class pulmonary disease classification.

S. Shahzad et al. [3] created an ensemble deep learning image model of robust multi-labelled chest focussed X-ray image classification, emphasizing the merits of multi-architectural constructions to increase the detection results of pneumonia and tuberculosis, which contributes to the validity of sophisticated deep models in pulmonary diagnosis.

S. K. Kabi et al. [4] proposes a new method of detecting tuberculosis and pneumonia based on the image of the chest X-ray in smart healthcare applications formed a solid background to automated pulmonary disease screening systems like the proposed one.

I. A. Ahmed et al. [5] Combined several analytical methods is conceptually related to our attention-enhanced feature representation, examined hybrid feature-based methods in the early detection of pneumonia and TB, which is why it is important to combine multiple methods of analysis.

M. S. Ahmed et al. [6] proposed a joint deep learning-based diagnosis of pneumonia, COVID-19, and TB with the help of chest focussed X-ray images, confirming the usefulness of co-disease classification models in practice.

V. Pal et al. [7] used transfer learning and deep cnn's to diagnose multifaceted diseases using chest X-rays, and the authors showed that pretrained architectures could enhance the generalization ability, which is the reason why our work will be based on transformer-based architectures.

P. Guddati et al. [8] applied a DCNN-based tuberculosis and pneumonia recognition framework to the FPGA platform, pointing to the possibility of real-time and resource-efficient implementation, which is also applicable to the clinical adaptation of our system in the future.

M. Vardhana et al. [9] introduced to pulmonary disease detection is focused on computational performance and embedded implementation, which offers the understanding of how deep learning models can be optimized to fit into a practical healthcare setting.

M. S. Al Reshan et al. [10] applied the MobileNet architecture to classify pneumonia based on the images of the chest X-ray that showed that lightweight deep learning models can be effectively applied to medical imaging tasks and provide a comparative baseline to transformer-based approaches.

A. Tripathi et al. [11] refined the classification of lung diseases with a better MobileNetV2 architecture demonstrating how the architecture could be refined to achieve a substantial improvement in early detection accuracy, which justifies the continuous architectural additions undertaken in our model.

A. A. Jeny et al. [12] Enhanced the relevance of ensemble and deep feature learning strategies to the analysis of medical images and created an eye disease detection and classification ensemble model, which conceptually is similar to multi-condition diagnosis systems.

D. S. Kermany et al. [13] proposed that image-based deep learning has the potential to recognize medical diagnoses and treatable diseases, which are the building blocks of the transformative power of deep neural networks in automated clinical decision support systems.

X. L. Yang and S. L. Yi [14] presented a multi-classification scheme of fundus disease detection based on DSRA-CNN, in which attention-based convolutional techniques are used, and this idea is conceptually analogous to the attention mechanisms that our Swin Transformer-based architecture incorporates.

III. Proposed System

The anticipated result of the suggested system is to create a practical deep learning-based algorithm to detect major respiratory illness in the initial stages with the help of chest X-ray photographs. In contrast to the old ones, single-disease classifiers, the present architecture is intended to classify three main categories (Normal, Pneumonia, and Tuberculosis (TB)) and can also be used to assess the severity of the lung infection to estimate the

level of lung infection. The system is designed to support the differences in the radiographic pattern and overlapping features and differentiate infectious cases and normal ones with the help of the system.

The base model uses Swin Transformer architecture as the backbone network which is upgraded with built-in attention mechanism to enhance feature representation and localisation of pathology specific lung regions. This improvement allows the network to selectively concentrate on the clinically significant pulmonary regions and the irrelevant background features can be avoided.

A. Data Collection:

To validate and train the featured artificial intelligence model, the chest X-ray medical images were acquired using datasets that were publicly available where the labelled examples of the cases of Normal, Pneumonia, and Tuberculosis could be found. The dataset consists of the images that were taken in various healthcare facilities and screening settings, and there is a variety in the demographics of the patients, imaging equipment, and different stages of the disease, which enhances the strength of the model and the models practical use in the real modern world.

To enhance the variety of the training data, the chest X-ray dataset was further structured so that there is equal representation of the Normal, Pneumonia, and Tuberculosis (TB) cases, and further augmentation measures were implemented so that the sample space could be enlarged. This strategy increased the quantity of training examples and allowed the model to acquire disease-related radiographic features as opposed to using dataset-level biases. The pictures are accompanied by one diagnostic category and severity data, which is a reflection of the real-world clinical practice where the scope of infection is different in patients. Such systematic labelling renders the dataset highly appropriate to the creation of a system that can both classify and approximate the severity. The chest focussed X-ray images are very variable in terms of contrast, illumination, anatomy positioning, and distribution of infections, which is similar to the variability experience in the routine radiology practice. This heterogeneity helps in enhancing the externalization ability of the model to various healthcare environments. The high-quality images were verified and organized, which only led to several thousand, which is an adequate and large enough base to train a strong attention-enhanced deep learning framework.

B. Data preprocessing:

The preprocessing phase starts with the loading and sorting of the data file directories of the labelled enhanced X-ray chest images. A stratified splitting strategy is used to make sure that the distribution of classes is balanced between the training and validation partitions, by setting 80 percent of the data to be used in training and 20 percent validation. An image path check assures that image paths are verified, corrupted or missing files are eliminated and that the referencing of files is consistent. The training data are augmented with the help of such techniques as resizing to 224x224 or 384x384 (depending on the model configuration), random horizontal flips, small rotations, and intensity normalization to improve the strength of the downstream model used. The validation data is only resized and normalized and not augmented, to ensure consistency of evaluation. Lastly, the datasets are turned into a set of tens and normalized by ImageNet means and standard deviation values so that they can be used with the pretrained Swin Transformer backbone. Moreover, the class weights are calculated using the distribution of training labels with an aim of reducing the class imbalance in optimizing the losses and achieving better general predictive performance.

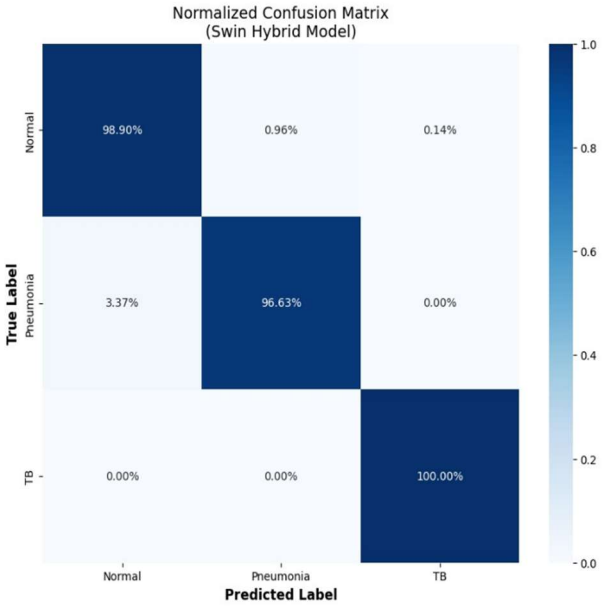

Fig. 1. Confusion Matrix

C. Model Training:

The modelling training was conducted based on the PyTorch Lightning framework that eases the training process and effectively balances hardware resources. A Swin Transformer backbone was designed as an architecture to classify chest X-rays by multi-class and specify the types of results (Normal, Pneumonia, and Tuberculosis) and severity at the same time. To deal with the identified imbalance of classes during the exploration

of the data, pre-calculated class weights were added to the model setup. The Trainer was set to apply 30-epochs training process with one GPU of 16-bit mixed-precision to speed up computation and optimize the use of memory.

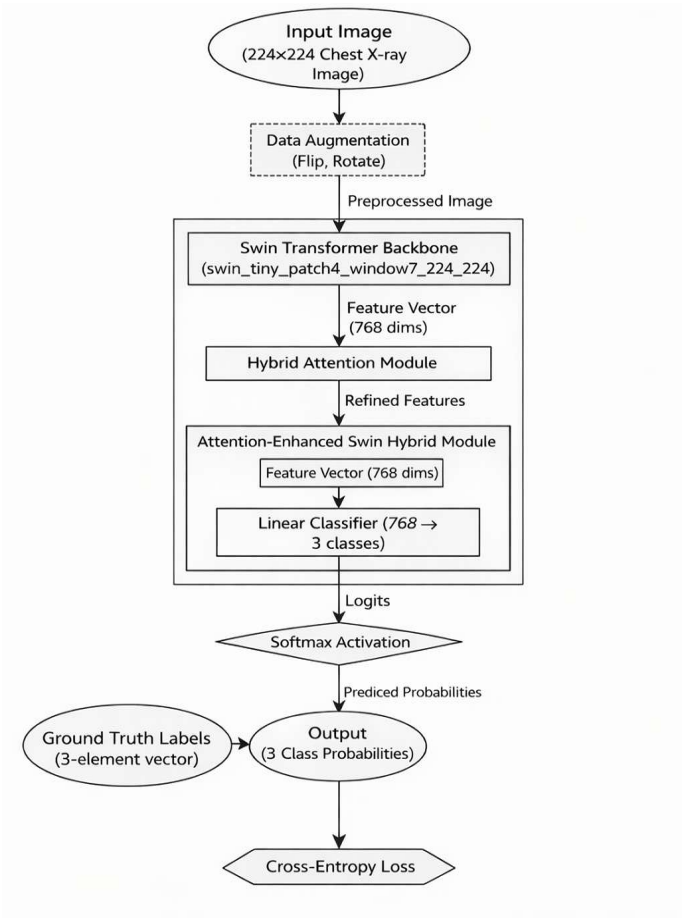

Fig. 2. Model Architecture

AdamW optimizer was used to update model parameters, and this is especially effective with transformer-based architectures. To manage the learning rate dynamics, a Cosine Annealing scheduler was applied where the learning rate decreased after following a sinusoidal pattern of decay across training epochs to allow the scheduler to converge more easily to a lower loss value. Cross-Entropy Loss was the loss function implemented which is the main loss function of multi-class classification with class weights to ensure that minority-class misclassification is more strongly penalized and that the learning is balanced. The training process was carried out by calling trainer.fit using the specified training and validation data loaders.

The model validation and working were measured using recall and precision, and also with f1-score and accuracy at the end of both training and validation data. The training logs showed a steady increase in performance throughout the 30 epochs, where the validation accuracy was near to high, and the F1-scores were high in the last epoch, which shows the model can correctly label the enhanced X-ray chest images and determine the severity of the disease with low overfitting.

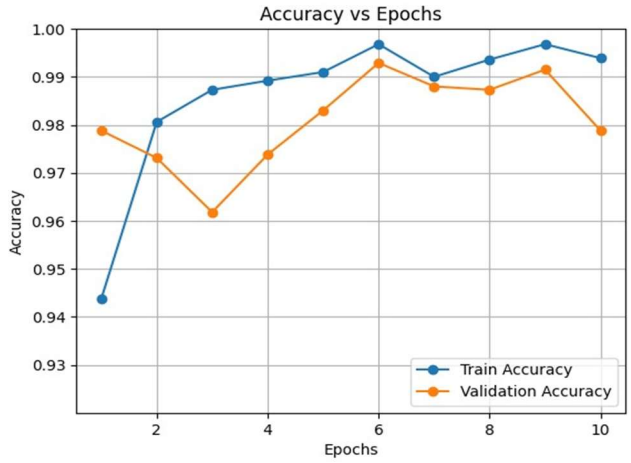

Fig. 3. Recall vs Epochs Graph

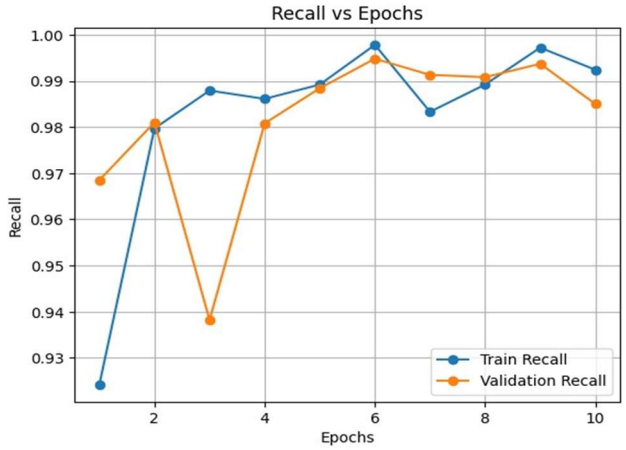

Fig. 4. Recall vs Epochs Graph

D. Advantages of proposed systems:

The main task of the proposed imaging system architecture based on a Swin Transformer architecture and extended with built-in attention mechanisms is to ensure that the diagnostic ability is much higher than with traditional solutions. The architecture reached a validation accuracy of about 98% which is much higher than the validation accuracy of traditional models of convolutional neural networks like the ResNet models and EfficientNet-based models that served as baselines. Thus, the given system proves to be a significant improvement to the conventional CNN-based methods of automated classification and severity of chest X-rays.

This can be explained by the fact that it has had a fundamental change in network architecture. In contrast to the traditional CNNs, which use localized convolutional operations with small receptive fields, Swin Transformer uses a hierarchical self-attention mechanism that uses shifted windows. This design allows both the co-occurrence of local lung abnormality of fine grain as well as the global contextual patterns that are evident throughout the whole chest X-ray image. This ability is specifically valuable in detecting small areas of infection and distinguishing between Pneumonia and Tuberculosis, which will allow carrying out a more detailed radiographic analysis.

Moreover, there is the incorporation of attention mechanisms which also contributes to improved performance since the network is able to highlight the most informative feature representations to pathological lung regions. This architectural improvement, along with a well-thought-out training plan (to solve the class imbalance and introduce the severity estimation) justifies the high-performance improvements seen in comparison to the previous CNN-based models.

E. Model deployment:

An implementation of the proposed Swin Transformer-based system will involve transferring the trained model out of the development environment into a clinically useful tool. The implementation plan will consist of incorporating the saved model checkpoint into a user-friendly computer interface where radiologists and other medical workers will be able to upload the images of the chest enhanced X-ray and get the immediate classification results and the severity assessment outputs.

The deployment plan gives reliability, availability and scalability priority as a way of guaranteeing successful performance in actual health care environments. The large validation accuracy and predictive consistency indicated by the high F1-score show how robust the system is. Moreover, the implemented solution should comply with the ethical principles and regulations to guarantee patient information privacy, safe management of medical data, and reliability among clinical users.

IV. Results And Discussion

The proposed system employs a Swin Transformer backbone with attention to the task of multi-classification of the chest focussed X-ray images into the fields of Normal, TB, and Pneumonia and determining the severity level. The training process was executed based on the PyTorch Lightning framework with 30 epochs in a computer with a graphics card of 16-bit mixed-precision arithmetic to guarantee that the computation is efficient and the memory usage is optimized. AdamW optimizer was used along with Cosine Annealing learning-rate scheduler in order to train the model parameters. Class imbalance in the data led to the adoption of stratified data splitting strategy and weighted Cross-Entropy loss was employed in the optimization process to favor minority-class misclassification and enhance balanced learning.

The model developed obtained a final validation accuracy of around 98% and a high F1-score, which is much higher than the earlier experiments that were carried out using the conventional convolutional neural networks like ResNet and EfficientNet-based models. Comparative analysis proved that there were visible performance improvements in the metrics of classification consistency and generalization ability. Confusion matrixes also helped to verify the strength of the system as the true-positive rates and false negatives were high in all types of diseases, which proved that the system is able to detect both healthy and infected lung cases effectively.

This high performance can be explained by the fact that the Swin Transformer is able to learn the global contextual dependencies by its hierarchical self-attention mechanism as opposed to the CNN architectures which are mainly based on local receptive fields. To be useful in deployment, the trained model is stored in a checkpoint file that can be reloaded to perform inference on previously unknown chest X-ray images. This would be implemented practically in encapsulation of the model in a web-based service, possibly based on a framework like FastAPI, to have a means of uploading images, applying the necessary preprocessing transformations, and returning structured classification and severity predictions. The solution is containerizable with Docker and can be deployed on a cloud platform in the clinical settings, guaranteeing scalability, reliability, and available automated lung disease screening.

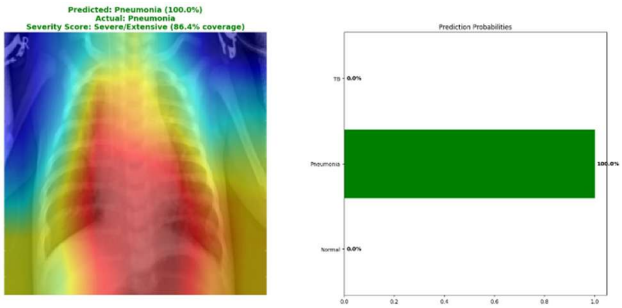

Fig. 5. Chest x-ray image diagnosis report

V. Conclusion And Future Works

The research has led to designing and validating a high-performance deep learning model of multi-class classification of major respiratory diseases with the help of chest X-ray images. The system had outstanding validation accuracy of about 98 percent and a robust F1-score through the use of Swin Transformer architecture with attention mechanisms and a severity assessing element. This approach is successful because the model has the capacity to represent complex contextual representations across the globe and localized pathological representations, paired with a well-developed training plan which includes addressing the issue of dataset imbalance using weighted loss functions and stratified data splitting. These results highlight the high potential of vision transformer architectures in the medical imaging field and form a solid basis on the creation of an automated system of lung disease screening and diagnostic assistance.

The future studies will aim at changing this high-performing model into a working clinical decision-support tool and enhancing its strength. The use of explainable AI methods, such as Grad-CAM, will be incorporated to improve interpretability and visualization of the lung areas that have the greatest impact on model predictions. Moreover, the system can be trained on larger, multi-institutional chest X-ray datasets to scale the system and enhance generalization to vary populations and imaging environments and increase the ability to correctly measure the severity of the disease in real-world conditions.

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